



A Visual Analytics Approach for Analysing Football Match Dynamics Using Positional Tracking Data

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A dissertation submitted in partial fulfilment of the
requirements for **MSc (in Data Science and Analytics)**
by taught programme, supervised by **Oktay Karakus**

APPENDIX 1: Specimen Layout for Declaration/Statements page to be included in Taught Master's Degree Dissertations

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Executive Summary

Football analytics has gained significant importance in modern sports performance analysis, driven by the increasing availability of positional tracking technologies and extensive football datasets. Traditional statistics like goals, possession, and pass counts offer only limited insights into team organisation and tactical strategies during matches. Conversely, positional tracking data captures ongoing player and ball movements, allowing for in-depth analysis of spatial arrangements, player positions, and tactical patterns. Despite their benefits, tracking datasets are highly complex and require advanced analytical and visualisation tools to interpret effectively.

This dissertation aimed to analyse football match dynamics by using positional tracking data, spatial metrics, and visual analytics techniques. It concentrated on processing tracking datasets to explore team organisation, spatial occupation, and tactical behaviour during matches. The study also sought to enhance the interpretation of complex tracking data through clear visualisations and comparative tactical analysis. The research particularly focused on understanding how spatial metrics could support tactical interpretation of team organisation and movement during different phases of play.

To meet these objectives, datasets from three football matches—Netherlands vs Senegal, France vs Argentina, and Germany vs Spain—were analysed. These datasets included timestamped player position coordinates recorded continuously during the games. The data were then pre-processed and analysed using Python libraries such as Pandas, NumPy, SciPy, Plotly, and Streamlit. Preprocessing procedures were used to organise continuous spatiotemporal tracking information into structured analytical outputs suitable for metric computation and visual exploration.

Several spatial metrics were calculated from the tracking data to analyse team structure and tactical behaviour. These included team width, team depth, compactness, centroid position, and convex hull area. Team width and depth assessed horizontal and vertical spread, while compactness measured player spacing around the team centroid. The convex hull area was used to evaluate the overall spatial occupation and team shape during various phases of play.

The study developed an interactive football analytics platform that visualises and compares spatial metrics using temporal line charts, pitch tactical views, comparative graphs, and summary metrics. These visualisations enable analysis of team movement, spatial organisation, and tactical changes throughout matches. Comparing data from multiple matches helps evaluate tactical differences between teams across different gameplay scenarios.

The results confirmed that positional tracking data provides valuable insights into tactical behaviour on the football pitch, which cannot be deduced solely from match events. Teams continuously adjusted their spatial formations during both offence and defence. Greater width and higher convex hull values indicated expanded pitch coverage and attacking opportunities, while lower compactness and reduced space utilisation reflected well-organised defensive setups. Additionally, variations were observed in the horizontal distribution, vertical arrangement, and teamwork throughout the matches studied. These findings demonstrated that team formations and spatial structures continuously evolved in response to attacking transitions, defensive organisation, and positional adjustments during gameplay.

The study emphasised the crucial role of processing and visualising tracking data in football analytics. Using spatial metrics alongside visual analytics techniques enhanced the interpretation of complex spatiotemporal data, leading to a clearer understanding of tactical structures and team behaviour. Combining various spatial metrics into a single analytical framework allowed for a more thorough tactical assessment than relying on isolated statistics.

While the study effectively analysed football positional tracking data and tactical behaviour, it had several limitations. It was confined to only three matches and mainly focused on team-level spatial analysis. Future research could expand by using larger datasets, event-based data, player-specific analysis, and machine learning methods to develop advanced tactical models and real-time football analytics.

Overall, this dissertation highlights the importance of positional tracking analysis and spatial metrics in contemporary football analytics. It illustrates how processing tracking data and employing visual analytics techniques can enhance understanding of match dynamics, team organisation, and tactical behaviour.

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Chapter 1: Introduction

1.1 Introduction

The use of data analytics in football has grown considerably with the advent of positional tracking technologies and sophisticated visualisation tools (Gudmundsson and Horton 2017). Tracking data offers in-depth insights into player movements, team formations, and spatial behaviour during matches, facilitating more thorough tactical analysis and performance assessment.

This dissertation focuses on developing an interactive visual analytics dashboard to analyse football match dynamics using positional tracking data. The project processes football positional tracking data to compute spatial metrics and extract tactical insights through interactive visual analytics techniques (Perin et al. 2013).

The dashboard was built using Python technologies, including Streamlit, Plotly, Pandas, and SciPy. It features metric visualisation, player movement analysis, tactical insights, and cross-match comparisons—tools designed to enhance interactive football data analysis and the interpretation of tracking data (Stein et al. 2018a).

1.2 Problem Statement

Modern football matches generate large volumes of positional tracking data that contain valuable information regarding player movement, team organisation, and tactical behaviour. Advances in tracking technologies have enabled clubs and analysts to capture detailed spatial-temporal data throughout matches, creating new opportunities for performance and tactical analysis (Gudmundsson and Horton 2017). However, analysing raw tracking data remains challenging due to its complexity, scale, and continuous nature.

Traditional football analysis methods primarily emphasise event-based statistics, including passes, shots, and possession, but offer limited insight into spatial team behaviour and tactical structures (Rein and Memmert 2016). Furthermore, most existing analytical systems lack interactive visualisation capabilities that enable users to dynamically explore positional data and tactical patterns during matches.

While spatial metrics like team width, compactness, centroids, and occupied area are common in football analytics research, effective interpretation depends on appropriate

visual analytics tools (Frencken et al. 2011). Without intuitive visualisation systems, it is challenging to detect tactical trends, compare team formations, and analyse player movement behaviours across various match scenarios. Processing positional tracking data and transforming it into interpretable tactical insights remain major challenges in modern football analytics.

Therefore, an interactive visual analytics system is needed to process positional tracking data, calculate relevant spatial metrics, and display tactical insights through user-friendly visualisations. This dissertation tackles this challenge by developing an interactive football analytics dashboard that analyses match dynamics using positional tracking data.

1.3 Aim of the Study

This dissertation aims to develop an interactive football visual analytics system that processes positional tracking data, computes spatial metrics, and supports tactical analysis through interactive visualisation techniques (Perin et al. 2013). It concentrates on calculating spatial football metrics and utilising interactive visualisation techniques to facilitate tactical analysis, team comparisons, and the examination of player movements across various matches (Clemente et al. 2014).

1.4 Objectives of the Study

The objectives of this study are:

1. To prepare and analyse football positional tracking data (Linke et al. 2020).
2. To compute spatial football metrics, including team width, team depth, compactness, convex hull area, and team centroid positions (Duarte et al. 2012).
3. To develop an interactive football analytics dashboard using Streamlit and Plotly (Perin et al. 2013).
4. To visualise player movement, team structure, and tactical patterns through interactive pitch visualisations and graphical analysis (Yue et al. 2008).
5. To carry out a comparative analysis of several football matches using calculated tactical metrics (CarlosLago-Peñas 2010).

6. To assess how interactive visual analytics can enhance the interpretation of football tracking data and aid tactical analysis (Stein et al. 2018a).

Chapter 2: Literature Review

2.1 Introduction

Progress in positional tracking technologies and extensive football datasets has elevated the role of football analytics in sports performance analysis. These innovations allow researchers to analyse player movement, tactical setups, and team dynamics more efficiently than traditional event-based statistics (Gudmundsson and Horton 2017).

This chapter surveys the literature on football positional tracking data, spatial metrics, data visualisation, and interactive dashboard systems. It reviews past studies on tactical analysis and spatial football analytics, highlighting research gaps that justify the creation of the proposed interactive football analytics dashboard.

2.2 Football Analytics

Football analysis refers to the use of statistics and data-driven models to assess players' performances and decisions, as well as tactical behaviour in football. It has increasingly been used over the years due to the development in technology used in gathering data and the need for evidence-based football analysis (Gudmundsson and Horton 2017).

Traditionally, football analyses have focused on event-based measurements, including goal-scoring rates, passing accuracy, tackling success, and possession percentage. While these measures offer important information about players' performances, they ignore spatial factors and game tactics (Rein and Memmert 2016). For this reason, current trends in football analysis have adopted spatiotemporal analysis using positional tracking data.

Various studies on football analytics have used game analysis methods to analyse tactics, movement patterns, pressing techniques, and territorial control (Link et al. 2016). Such analyses provide insights into understanding the collective tactical behaviour and organisation of teams at different stages of the game.

Advances in computational processing have further improved the analysis of complex football datasets (Stein et al. 2018b). Consequently, football analytics has become increasingly important in modern tactical analysis and performance evaluation.

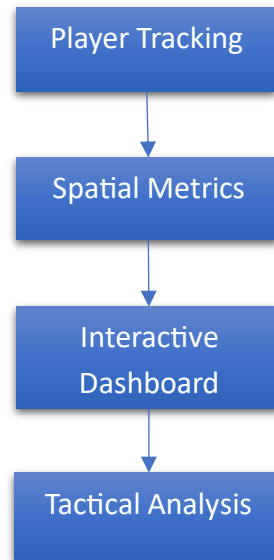


Figure 1: Overview of Football Analytics Workflow

Figure 1 depicts the typical workflow in football analytics systems, starting with the collection of positional tracking data and concluding with the generation of tactical insights through visual analysis.

2.3 Positional Tracking Data in Football

Positional tracking data is now a key part of modern football analysis, regularly capturing players' and the ball's spatial positions throughout matches. Unlike traditional event-based statistics, these tracking systems enable real-time, detailed analysis of movement patterns, team formations, and tactical behaviours (Gudmundsson and Horton 2017).

Technologies such as GPS systems, optical cameras, and wearable sensors are commonly used in professional football to capture high-frequency spatiotemporal data (Linke et al. 2018). These systems generate large positional datasets that support the analysis of player movement, pressing behaviour, defensive organisation, and team coordination.

Tracking data has greatly enhanced football tactical analysis by allowing researchers to better analyse spatial positioning, formation shifts, and player interactions compared to traditional match statistics (Memmert et al. 2017). However, these datasets tend to be high-dimensional and require significant computational power, making preprocessing, filtering, and visualisation essential for meaningful analysis (Stein et al. 2018b). As a result, interactive dashboards and visual analytics systems have gained importance for converting complex tracking data into clear tactical insights.

2.4 Spatial Metrics in Football

Spatial metrics are crucial in modern football analysis because they offer quantitative insights into how teams organise and execute tactics. Unlike conventional stats like possession or pass counts, these metrics analyse how players collaboratively fill and control space during various phases of play (Frencken et al. 2011).

Tactical effectiveness in football relies greatly on player positioning, team spacing, defensive compactness, attacking width, and positional coordination. Spatial metrics convert positional tracking data into numerical indicators that depict team structure and movement patterns (Clemente et al. 2015).

Researchers often evaluate horizontal and vertical player distribution using metrics like team width and depth. Compactness assesses how close players are to the team centroid, while convex hull area measures the total pitch space covered by the team (Duarte et al. 2012).

The growing availability of high-frequency tracking data has made spatial metrics increasingly important in football research. These methods support tactical analysis by detecting formation changes, coordinating movements, and assessing spatial control during matches (Ric et al. 2017).

2.4.1 Team Width

Team width indicates how players are spread across the football pitch horizontally during a match. It is typically measured as the distance between the leftmost and rightmost players at any given moment (Frencken et al. 2011). This metric is a key tactical indicator, as it shows how well a team uses lateral space during both attacking and defensive situations.

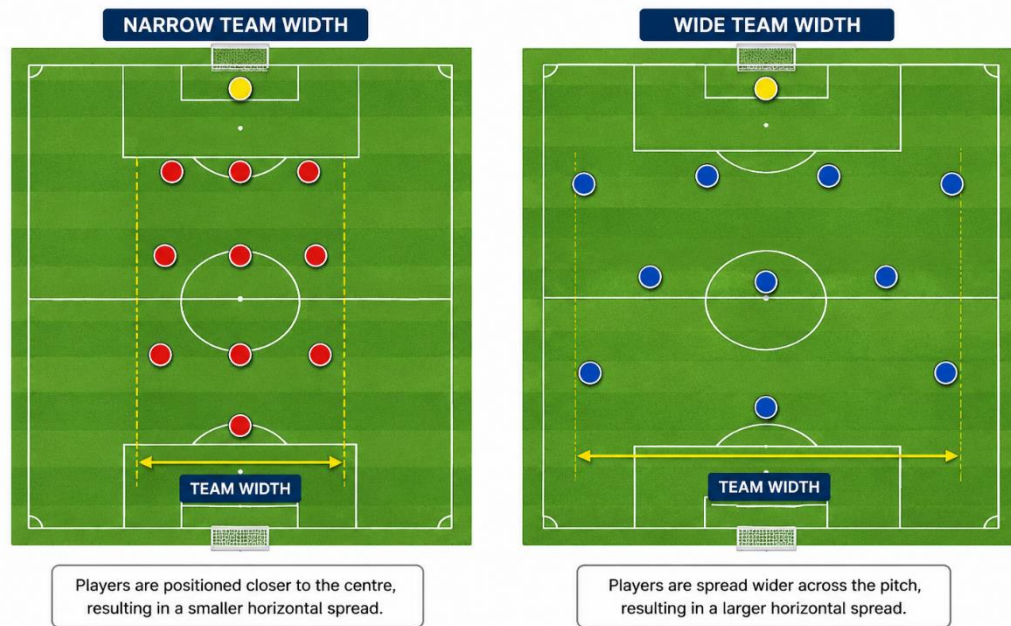


Figure 2: Illustration of Team Width in Football

In attacking scenarios, a wider team formation generally provides better spatial coverage and allows teams to stretch defensive lines horizontally (Silva et al. 2015). Such wide formations can open up passing options, enhance ball movement, and create gaps between the opposing defenders. Consequently, teams that successfully sustain their width tend to boost their attacking fluidity and positional adaptability.

In contrast, defensive phases usually feature a narrower team width to maintain a compact defensive shape and limit gaps opponents can exploit (Clemente 2014). Such narrow formations enable teams to exert collective pressure around the ball and restrict the attacking choices available to the opposition in central regions of the pitch.

Researchers have analysed how team width varies across different tactical systems, match situations, and phases of play using positional tracking data (Yue et al. 2008). These studies show that team width fluctuates throughout a football match based on possession, pressing intensity, and tactical approach. As a result, width analysis has become a common spatial metric for assessing football performance and tactics.

2.4.2 Team Depth

Team depth describes the vertical span between the most defensive and most advanced players on a football team during a match. It indicates how far the team extends from

the defensive line to the attacking front and is often used to assess tactical structure and spatial setup (Mackenzie and Cushion 2013).

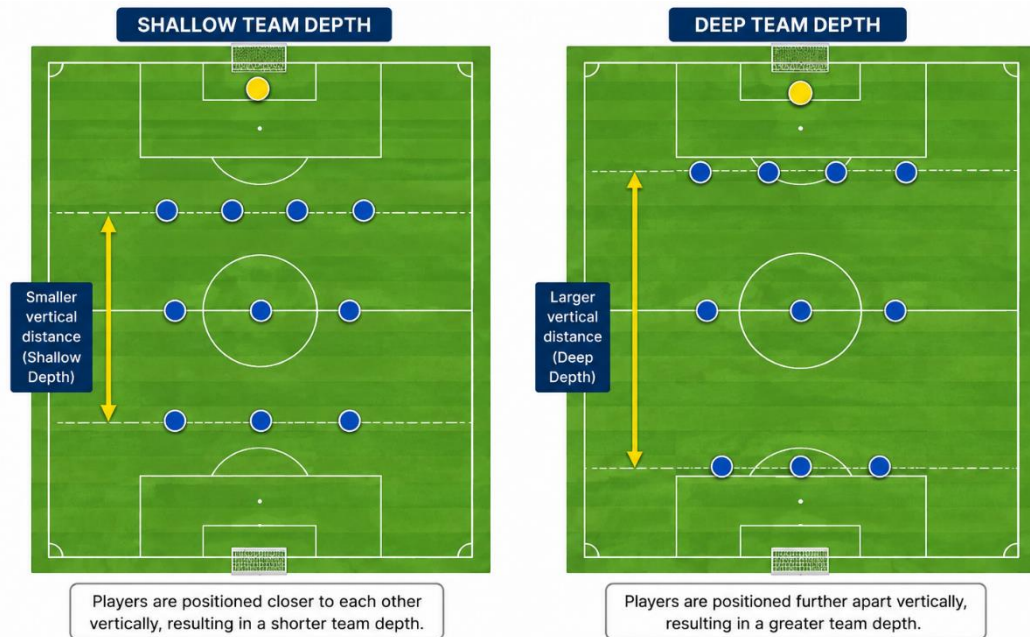


Figure 3: Illustration of Team Depth in Football

In attacking scenarios, having greater team depth can facilitate forward advancement and enable vertical passing options between defensive and offensive units. Teams that make good use of depth can reveal gaps behind the opposition's defensive lines and enhance attacking options through coordinated forward pushes.

However, excessive defensive depth can create larger gaps between the defensive and midfield units, decreasing overall compactness and making the team more vulnerable to opponent attacks (Duarte et al. 2013). Therefore, successful teams typically aim to maintain a balanced depth to ensure defensive stability while effectively supporting offensive transitions.

Research in football analytics indicates that team depth varies dynamically based on tactical strategy, match tempo, and possession. For instance, high-pressure teams often keep players closer vertically to enhance collective pressure and limit opponents' space. Conversely, counter-attacking teams might increase depth temporarily to take advantage of open attacking areas.

Positional tracking technologies allow researchers to continuously analyse team depth during football matches, offering deeper insights into tactical transitions and collective

movement patterns. Consequently, depth analysis has become a key metric for assessing modern football performance and tactics.

2.4.3 Team Compactness

Team compactness refers to how players are positioned close to each other during a football match. It is often linked to a team's capacity to stay coordinated, minimise exploitable space, and maintain tactical organisation in both attack and defence (Duarte et al. 2012).

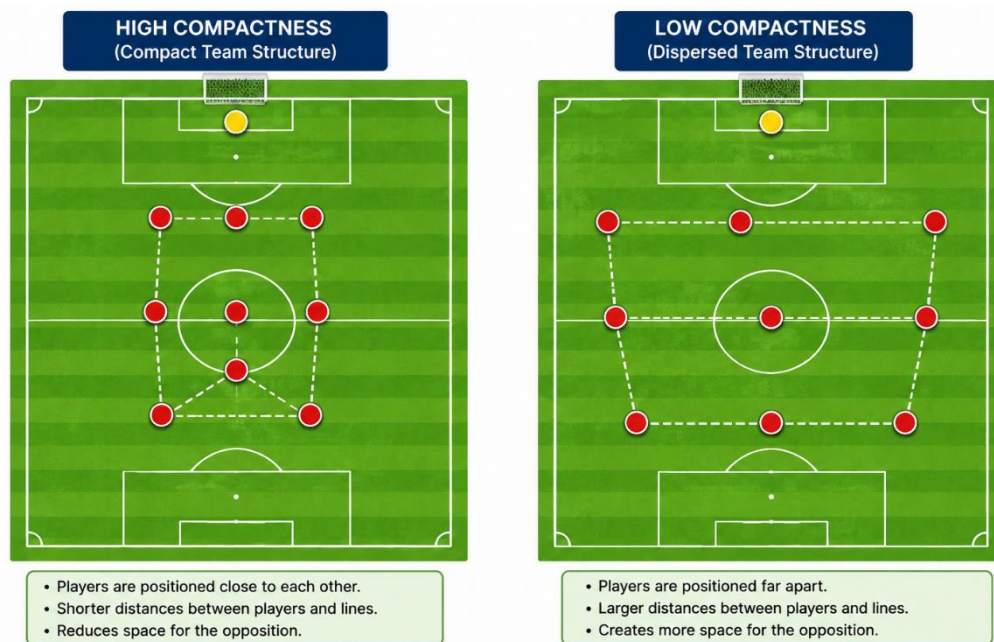


Figure 4: Illustration of Team Compactness in Football

In defensive scenarios, compact teams typically minimise space for opponents by keeping players and tactical units closer together (Silva et al. 2016). This organised arrangement enhances pressing, defensive coverage, and team-wide ball recovery. Teams with high defensive compactness tend to be better at blocking passing lanes and restricting central attacking areas.

Compactness is crucial during attacking transitions. Keeping players evenly spaced facilitates passing, supports proper positioning, and ensures smooth movement across the field. Large gaps between players can weaken tactical cohesion and hinder both ball movement and team coordination.

Multiple studies have used positional tracking data to assess compactness using centroid-based distances and spatial relationships among players (Frencken et al.

2011). These methods enable researchers to analyse how teams naturally expand and contract during various phases of play. Changes in compactness are often linked to tactical setups, pressing intensity, and the overall context of the match.

The growing adoption of tracking technologies allows for continuous monitoring of compactness metrics in football matches, enhancing insights into collective tactical behaviour and spatial organisation. As a result, compactness analysis has become an important part of football performance assessment and tactical evaluation systems.

2.4.4 Convex Hull Area

Convex hull area is a spatial metric in football analytics that measures the total space occupied by a team on the pitch at a given moment. It is calculated by constructing the smallest convex polygon that encompasses all player positions (Clemente et al. 2015). This analysis offers valuable insight into how teams collectively utilise and control space during a match.

Researchers often use convex hull metrics to analyse tactical expansion, defensive structure, and spatial control in football matches (Frencken et al. 2011). A larger convex hull area typically indicates players are more spread out across the pitch, whereas a smaller area indicates greater compactness and more cohesive tactical organisation.

During attacking phases, teams often increase their spatial coverage to create passing options and stretch the opposition both horizontally and vertically (Duarte et al. 2012). This broader spatial occupation supports ball circulation, positional flexibility, and attacking progression. Conversely, defensive phases usually involve reduced convex-hull area to maintain compactness and to restrict opponents' movement and passing opportunities.

Convex hull analysis has gained importance as positional tracking and computational systems advance. By tracking player positions throughout matches, analysts can assess shifts in team spatial occupation during transitions, pressing, and tactical changes.

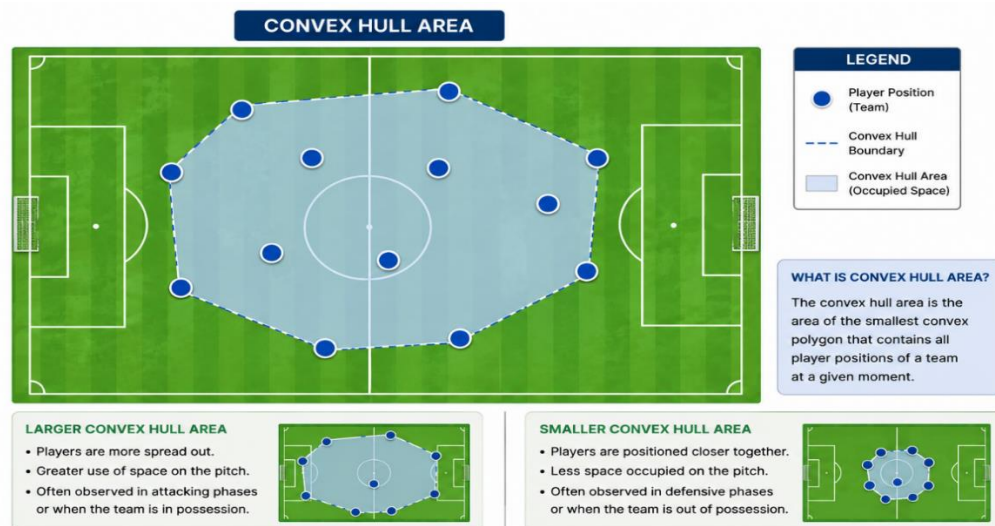


Figure 5: Illustration of Convex Hull Area in Football

Convex hull metrics are commonly utilised to analyse collective tactical behaviour and spatial organisation in football (Ric et al. 2017). As a result, the convex hull area has emerged as a key spatial metric in contemporary football analytics.

2.5 Data Visualisation in Football Analytics

Data visualisation is crucial in football analytics because it transforms complex positional and performance information into understandable visual formats (Gudmundsson and Horton 2017). Football tracking data records ongoing player movements, spatial positions, and tactical changes during a match, making clear and effective visualisation key to recognising tactical trends and behaviours.

Visualisation methods like pitch maps, heatmaps, temporal line charts, movement trails, and spatial polygons assist analysts in analysing team positioning, movement coordination, and tactical structure over time (Perin et al. 2013). These techniques are especially useful for positional tracking, as they maintain the spatial context of the football pitch and enhance the understanding of tactical behaviour.

Interactive visualisation enhances football analysis by allowing users to filter, compare, and explore match data dynamically (Stein et al. 2018b). In this dissertation, visualisation is embedded into the analytical process with line charts, bar charts, pitch visualisations, and convex hull overlays. These visual tools aid in interpreting spatial metrics and tactical behaviour by connecting numerical tracking data with interactive visual insights.

Consequently, data visualisation plays a crucial role in improving the accessibility and interpretation of football positional tracking data for tactical analysis.

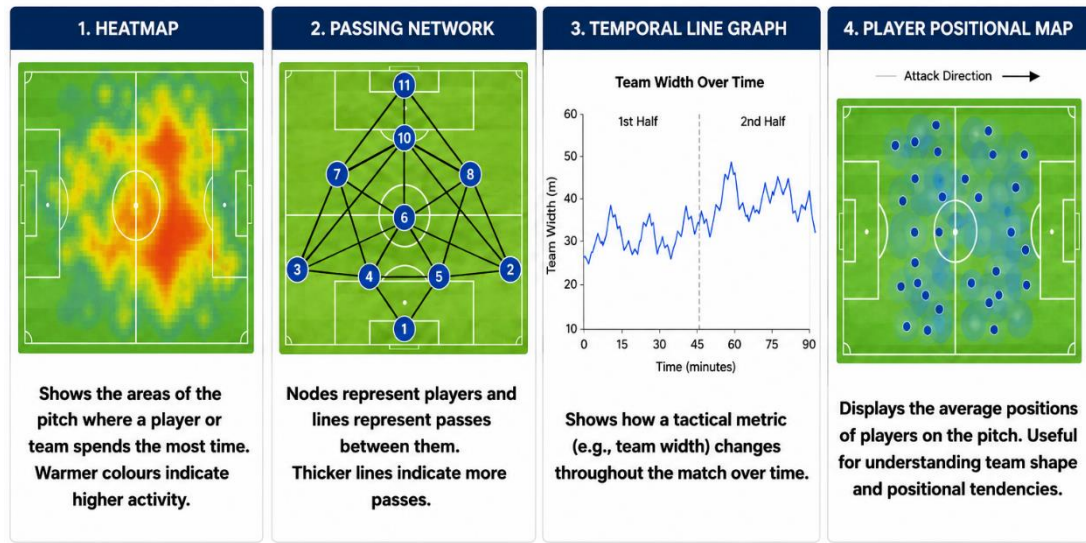


Figure 6: Common Visualisation Techniques Used in Football Analytics

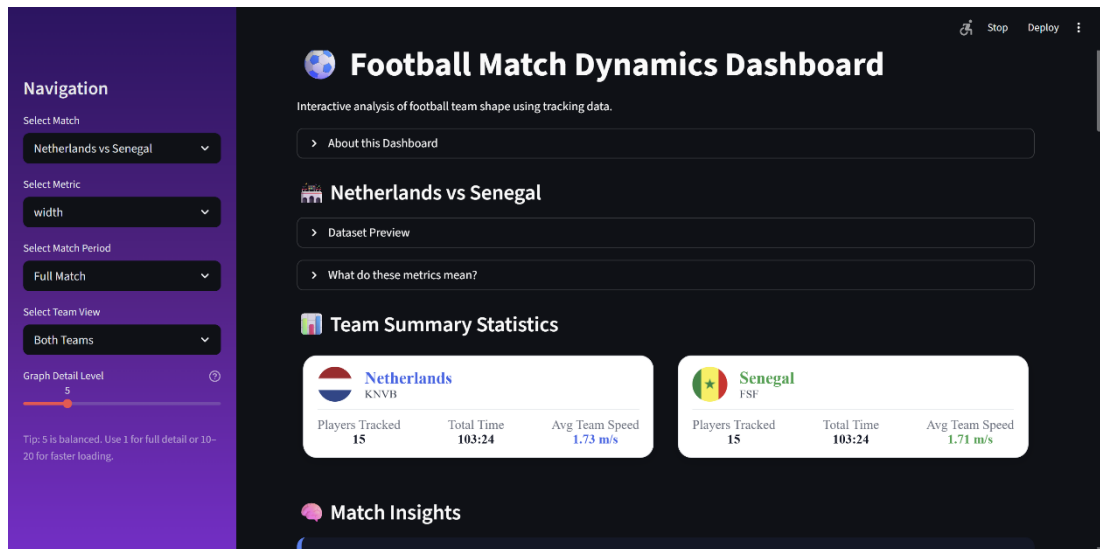
2.6 Interactive Dashboards for Football Analysis

Interactive dashboards are commonly utilised in football analytics to organise and analyse extensive match data through a single interface. By integrating charts, filters, and visual elements, these dashboards allow for more efficient engagement with football data than traditional static reports.

Football match data contains continuously changing spatial and temporal variables, making interactive exploration important for tactical analysis (Perin et al. 2013). Dashboards enable users to compare teams, analyse tactical metrics, review match periods, and explore positional changes in real-time during gameplay.

Modern football dashboards often include filters for matches, teams, and time periods, combined with visual tools like line charts, positional plots, heatmaps, and comparison graphs (Stein et al. 2018b). These features enhance understanding of team structure, spatial behaviour, and tactical organisation.

The dashboard developed in this dissertation uses Streamlit and Plotly to process positional tracking data and produce interactive tactical visualisations. The system enables spatial metric analysis, cross-match comparison, and interactive tactical exploration using dynamic filtering and visual analytics tools.



*Figure 7: Homepage of the Interactive Football Match Dynamics Dashboard
Developed in This Study*

2.7 Research Gap

Existing football analytics research has widely explored positional tracking systems, spatial metrics, and tactical performance analysis methods (Frencken et al. 2011). Previous studies have also investigated visual analytics approaches for analysing sports data and tactical behaviour (Perin et al. 2013).

However, many existing football analytics systems remain technically complex and primarily focus on statistical or algorithmic analysis rather than interactive tactical exploration (Stein et al. 2018b). Several studies also analyse spatial metrics individually rather than integrating multiple tactical metrics into a unified analytical environment.

Additionally, limited research has focused on interactive cross-match comparison and dynamic visual exploration of tactical behaviour across different gameplay situations (Fernández and Bornn 2018).

Despite the growing use of positional tracking data and spatial metrics in football analytics research, many existing studies primarily focus on isolated statistical analysis or highly technical modelling approaches. Limited attention has been given to integrating multiple spatial metrics, interactive visual exploration, and comparative tactical analysis within a single accessible analytical environment. Furthermore, several existing systems provide limited support for intuitive user interaction and

dynamic cross-match tactical comparison. This dissertation addresses these limitations by developing a football-focused visual analytics dashboard that combines positional tracking analysis, spatial metric computation, and interactive tactical visualisation within a unified platform. It also highlights usability and visual interpretation alongside the computational analysis.

2.8 Chapter Summary

The current chapter analysed previous studies on football analytics, including positional tracking, spatial data, data visualisation, and interactive dashboard applications. Previous studies have shown that tracking systems and spatial methods can be used to analyse football tactics and team performance in games.

The chapter also discussed key spatial metrics such as team width, depth, compactness, and convex hull area, highlighting their importance in analysing team organisation and spatial behaviour. It further analysed how data visualisation and interactive dashboards improve the interpretation of football tracking data.

Finally, the research highlighted gaps in existing football analytics, particularly in accessibility, the integration of diverse visual analytics techniques, and interactive tactical exploration. These insights serve as the foundation for the methodology and dashboard development outlined in the following chapter.

Chapter 3 — Methodology

3.1 Introduction

This chapter outlines the methodology for analysing football positional tracking datasets and analysing team spatial behaviour during matches. It details the research approach, dataset structure, preprocessing procedures, computation of spatial metrics, visualisation methods, and the implementation process used throughout the study.

The methodology mainly involves analysing positional tracking data to derive tactical insights via spatial and temporal analysis. Given the extensive volume of continuous spatiotemporal information in tracking datasets, preprocessing and metric extraction were crucial for converting raw positional data into meaningful indicators of team organisation and tactical behaviour. Python libraries and analytical tools facilitated data processing, computation of spatial metrics, and visual analysis.

This chapter details the implementation workflow used in the study, covering data preparation, frame-by-frame metric calculation, visualisation design, and how analytical components are integrated into the developed system. The methods outlined here provide the technical basis for the results and tactical analysis in the next chapter.

3.2 Research Methodology Overview

3.2.1 Research Approach

This study takes an analytical and implementation-oriented approach to analyse football match dynamics through positional tracking data. It merges spatial metric analysis with visual analytics techniques to explore team movement, spatial organisation, and tactical behaviour during matches.

The research mainly concentrates on analysing high-frequency positional tracking data and translating raw spatial coordinates into understandable tactical indicators. Python was utilised for preprocessing data, calculating spatial metrics, and creating visualisations. The analysis workflow aims to convert complex tracking datasets into straightforward visual formats that assist in tactical interpretation and enable comparative football analysis.

Spatial metrics such as team width, depth, compactness, centroid location, and convex hull area were calculated from player positional data. These metrics were then incorporated into line charts, pitch visualisations, summary tables, and comparative graphics to aid in analysing team formations and spatial patterns during matches.

3.2.2 System Workflow Overview

The study's workflow started with importing positional tracking datasets from three football matches: Netherlands vs Senegal, France vs Argentina, and Germany vs Spain. These datasets were analysed and organised using Python libraries such as Pandas and NumPy before further analysis.

After the initial data exploration stage, player positional coordinates were grouped by timestamp and team identifier to enable frame-by-frame spatial analysis. Spatial metrics, including centroid position, team width, depth, compactness, and convex hull area, were then computed using numerical and geometric methods.

The metrics were incorporated into an interactive dashboard built with Streamlit and Plotly, enabling users to analyse football match dynamics through interactive filters, visual comparisons, and tactical visualisations.

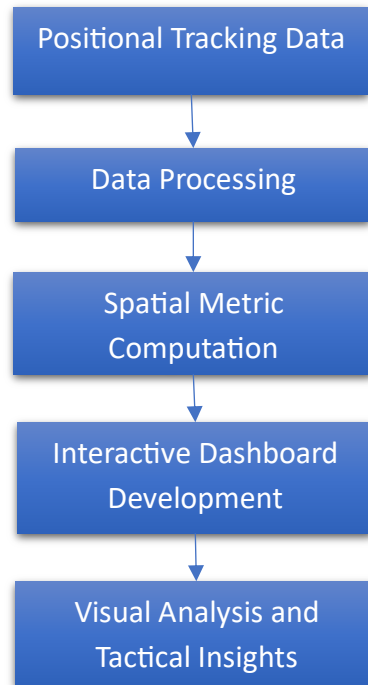


Figure 8: Overall Methodology Workflow of the Study

3.3 Dataset Description

3.3.1 Positional Tracking Dataset

This study used football positional tracking datasets that recorded player movements frame by frame during matches. The datasets continuously captured player positions throughout gameplay, enabling analysis of team behaviour both spatially and temporally across various phases of play.

Three football matches—Netherlands vs Senegal, France vs Argentina, and Germany vs Spain—were chosen for analysis. These matches served as the basis for calculating spatial metrics, visualising tactics, and conducting comparative evaluations within the interactive dashboard.

3.3.2 Dataset Features and Variables

The datasets included positional, temporal, and match-related variables essential for football analytics. These key variables comprised positional coordinates (x, y),

timestamps (timestamp_s), player IDs (player_id), team information (team_name), frame numbers (frame_num), and match periods (period).

The datasets also included variables such as ball_x, ball_y, pitch_length, and pitch_width. These positional coordinates served as the primary input to compute various spatial metrics, including team width, depth, compactness, centroid position, and convex hull area.

The frame-based structure of the tracking data enabled continuous observation of player movement and team positioning throughout the match. Figure 9 presents a sample of the positional tracking dataset from the Netherlands vs Senegal match, showing key variables used for spatial analysis and metric computation within the dashboard system.

	timestamp_s	team_name	player_name	x	y	speed	period
0	141.007674	Senegal	Youssouf Sabaly	13.975	19.244	0.179	1.0
1	141.007674	Senegal	Pape Abou Cissé	8.035	6.507	0.105	1.0
2	141.007674	Senegal	Edouard Mendy	41.257	-0.454	0.574	1.0
3	141.007674	Senegal	Nampalys Mendy	9.283	-0.842	0.425	1.0
4	141.007674	Senegal	Abdou Diallo	17.438	-22.039	0.156	1.0

Figure 9: Sample Positional Tracking Dataset Showing Key Variables Used for Analysis

3.3.3 Selected Matches for Analysis

The selected matches were chosen to support comparative analysis of different tactical behaviours and spatial structures in football matches. Using multiple datasets enabled the study to analyse variations in team organisation, spatial occupation, and metric patterns across different gameplay situations and tactical contexts.

The selected match datasets were further used to analyse team width, depth, compactness, and convex hull area through line charts, pitch visualisations, summary tables, and comparative graphical outputs. This comparative approach supported the evaluation of football match dynamics and spatial behaviour across different teams and matches.

3.4 Data Exploration and Preprocessing

3.4.1 Exploratory Data Analysis

Before computing metrics, Exploratory Data Analysis (EDA) was performed to understand the structure and features of the positional tracking datasets. Methods like `df.info()` and `df.describe()` were utilised to analyse variable types, dataset size, numerical distributions, and the overall dataset makeup.

The analysis verified that the datasets included positional coordinates, time-related variables, player details, and match attributes essential for spatial analysis and creating dashboards.

3.4.2 Team and Timestamp Analysis

Team-level exploration involved using `df['team_name'].unique()` and `groupby()` to identify participating teams and count the number of unique players per team.

Timestamp values were analysed through their minimum and maximum to determine the temporal span of the match recordings. This analysis helped clarify the frame-based organisation of the tracking datasets.

3.4.3 Missing Value Analysis

Missing value analysis was performed using `df.isnull().sum()` to detect incomplete variables in the datasets. While most player tracking variables had complete data, several ball-related variables—such as `ball_x`, `ball_y`, `ball_z`, `ball_x_plot`, `ball_y_plot`, and `ball_visibility`—had significant missing values.

Since this study mainly focused on player positioning and team spatial dynamics, the metric calculations primarily relied on player-tracking data rather than ball-related features. This ensured that the subsequent spatial metric calculations remained focused on reliable player positional information throughout the analysis.

3.4.4 Data Filtering and Transformation

The preprocessing stage included organising and filtering the datasets before calculating metrics. Teams and game periods filtered the match data to facilitate team-level spatial analysis.

Player positional coordinates (x, y) were organised by timestamps and team IDs to support frame-by-frame metric analysis. This organisation was used to analyse team width, depth, compactness, centroid positions, and convex hull area throughout the matches. Since positional tracking datasets include extensive continuous spatiotemporal data, preprocessing and metric extraction were vital for converting raw positional coordinates into meaningful indicators of team structure and tactical actions.

3.5 Spatial and Tactical Metrics

Spatial and tactical metrics were derived from positional tracking data to analyse team formation, field coverage, and player arrangement during football matches. These metrics were calculated frame-by-frame using player coordinates organised by timestamp and team ID. Python libraries such as Pandas, NumPy, and SciPy facilitated these calculations.

The spatial metrics were generated using a custom Python function named `compute_team_metrics()`. The grouped tracking data was processed using:

```
df.groupby(['timestamp_s', 'team_name'])
```

This grouping structure allowed for the calculation of team-level metrics at each timestamp during the match. These metrics quantified team organisation but did not fully capture tactical context.

3.5.1 Team Centroid Calculation

The team centroid indicates the average position of all players at a given timestamp. It is determined by calculating the mean of each player's positional coordinates.

```
centroid_x = group['x'].mean()  
centroid_y = group['y'].mean()
```

The centroid values were later used in compactness calculations and spatial visualisation of team movement across the pitch.

3.5.2 Team Width Calculation

Team width was calculated to measure the horizontal spread of players across the football pitch. The metric was computed as the difference between the maximum and minimum x coordinate values within a team.

$$\text{Width} = x_{\max} - x_{\min}$$

In Python, the calculation was implemented as:

```
width = group['x'].max() - group['x'].min()
```

Higher width values indicated wider team structures, while lower values represented narrower spatial organisation.

3.5.3 Team Depth Calculation

Team depth was calculated to analyse the vertical spread of players from defensive to attacking areas of the pitch. The metric was computed as the difference between the maximum and minimum y-coordinate values.

$$\text{Depth} = y_{\max} - y_{\min}$$

The corresponding Python implementation was:

```
depth = group['y'].max() - group['y'].min()
```

This metric was used to evaluate vertical team organisation and tactical spacing during matches.

3.5.4 Team Compactness Calculation

Compactness was used to measure how closely players were positioned relative to the team centroid. Euclidean distance was calculated between each player and the centroid position.

$$\text{Compactness} = \frac{1}{n} \sum_{i=1}^n d(p_i, c)$$

The distances were computed using:

```
distances = np.sqrt(
    (group['x'] - centroid_x) ** 2 +
    (group['y'] - centroid_y) ** 2
)
```

The mean of these distances represented the overall compactness of the team structure.

```
compactness = distances.mean()
```

Lower compactness values indicated that players were positioned closer together, while higher values represented greater spatial dispersion.

3.5.5 Convex Hull Area Calculation

Convex hull area was calculated to measure the overall spatial occupation of a team on the pitch. The convex hull was generated using the outermost player positions within a team.

The positional coordinates were first converted into coordinate points:

```
points = group[['x', 'y']].values
```

The convex hull area was then calculated using the ConvexHull() function from the SciPy spatial library:

```
hull = ConvexHull(points)
area = hull.volume
```

A minimum of three positional points was required to generate the convex hull polygon. If insufficient points were available, the area value was assigned as NaN.

The computed spatial metrics were stored in a new data frame and later integrated into the interactive dashboard for line chart visualisation, comparative analysis, and pitch-based tactical exploration. These metrics collectively enabled quantitative evaluation of team structure and spatial organisation during football matches.

	timestamp_s	team_name	centroid_x	centroid_y	width	depth	compactness	area
0	141.007674	Netherlands	-11.232727	-1.615545	28.596	58.158	16.204250	1070.333050
1	141.007674	Senegal	12.530273	-1.901091	41.293	41.283	14.738718	1082.244316
2	141.207875	Netherlands	-11.266818	-1.497909	29.524	58.154	16.243542	1080.751675
3	141.207875	Senegal	12.524727	-1.915182	41.568	41.199	14.746164	1084.732161
4	141.408075	Netherlands	-11.367636	-1.108000	31.038	58.301	16.293198	1092.489516

Figure 10: Sample Output of Computed Spatial Metrics

Figure 10 presents a sample of the computed spatial metrics generated from the positional tracking data. The output includes centroid positions, team width, depth, compactness, and convex hull area for each team at different timestamps during the match.

3.6 Dashboard Development

3.6.1 Development Environment

The interactive football analytics dashboard was developed using Python within Google Colab and Visual Studio Code environments. Google Collab was primarily used for data exploration, preprocessing, spatial metric computation, and initial visual analysis of the positional tracking datasets.

The final dashboard interface and application were developed using Visual Studio Code and Streamlit. The development workflow combined positional tracking data processing, spatial metric calculations, and interactive visualisation components into a single analytical dashboard.

The development environment supported data handling, graphical visualisation, and interactive analytical controls for positional tracking analysis.

3.6.2 Python Libraries and Tools

Several Python libraries were used throughout the development of the dashboard system. Pandas and NumPy were employed for data manipulation, filtering, grouping, and numerical computation. SciPy was used for geometric calculations, particularly to compute the area of the convex hull using the `ConvexHull()` function.

Plotly was used to generate interactive visualisations, including line graphs, bar charts, scatter plots, and pitch-based visualisations. Streamlit was used as the primary framework for building the dashboard interface and integrating interactive components.

The combination of these libraries enabled efficient processing of positional tracking data and supported the development of an interactive visual analytics platform for football match analysis.

3.6.3 Streamlit Dashboard Design

The system interface was developed using Streamlit to support analysis of football positional tracking data and spatial metrics through an organised analytical layout. The interface included a sidebar navigation panel, metric selection controls, match filters, and sections for tactical and statistical outputs.

Match selection features, dataset preview, team statistics data, as well as graphics generated from positional tracking analysis, were all available on the homepage. This structure allowed for both the analysis of individual matches and the comparison of several football matches.

Different visualisations have been employed within the system to study different attributes such as width, depth, compactness, space occupation, and tactical organisation, among others. These visualisations included temporal line charts, pitch-based positional charts, bar charts, table summaries, and convex hull visualisations. Emphasis was placed on designing the interface in such a way that it was easily readable.

3.6.4 Interactive Components and Filters

Filtering components were incorporated into the system to improve usability and analytical flexibility. Users can select different matches, teams, metrics, and match periods using the dropdown filters provided within the sidebar navigation panel.

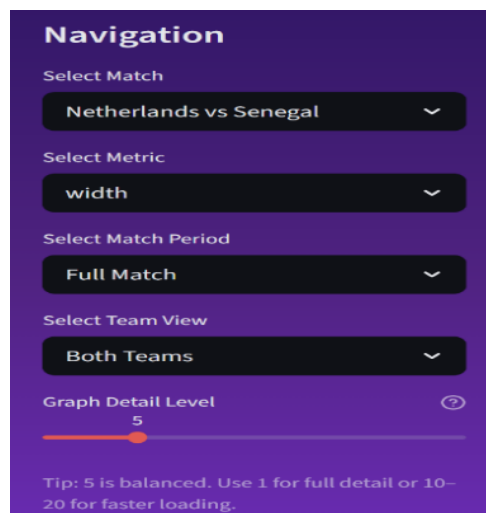


Figure 11: Interactive Dashboard Filters and Navigation Controls

A graph details slider has also been provided in this system, which will help users to maintain the level of detail versus performance trade-off while analysing positional tracking information. Filters have been provided to analyse differences in metrics, positional behaviours, and tactics.

Additional interface elements such as expandable dataset previews, metric explanation panels, and comparative graphical outputs were added to enhance accessibility and facilitate exploration of football positional tracking data.

3.7 Data Visualisation Techniques

3.7.1 Pitch-Based Visualisation

Pitch-based visualisations were created to directly display player positions and team spatial arrangements on a football pitch layout. They utilise (x, y) positional coordinates from tracking data to illustrate player distributions and tactical setups during various match scenarios.

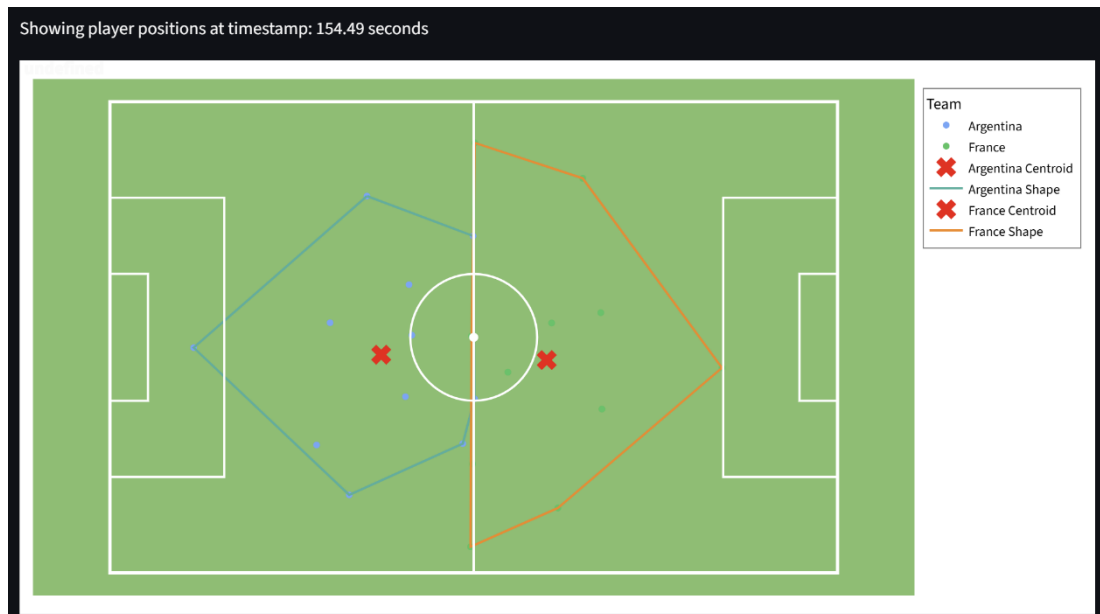


Figure 12: Pitch-Based Positional Visualisation from the France vs Argentina Match

Figure 12 shows a pitch-based positional visualisation from the France vs Argentina match at a selected timestamp. The figure displays player positions, team centroid locations, and convex hull structures generated from positional tracking coordinates.

Scatter-based pitch plots were added to the dashboard to show player positions dynamically across the field. These visualisations allowed for analysis of team formation, spatial coverage, and player positioning during the game. Additionally, positional plots were combined with convex hulls to depict the effective area occupied by teams at various moments.

Interactive pitch visualisations supported tactical interpretation by transforming complex positional tracking data into intuitive spatial representations. These visual outputs facilitated frame-by-frame spatial analysis and enabled comparative tactical exploration within the dashboard environment.

3.7.2 Temporal Line Charts

Temporal line charts were created to analyse how spatial metrics evolved during football matches. These charts displayed continuous variations in metrics such as team width, depth, compactness, and convex hull area using timestamp-based positional tracking data.

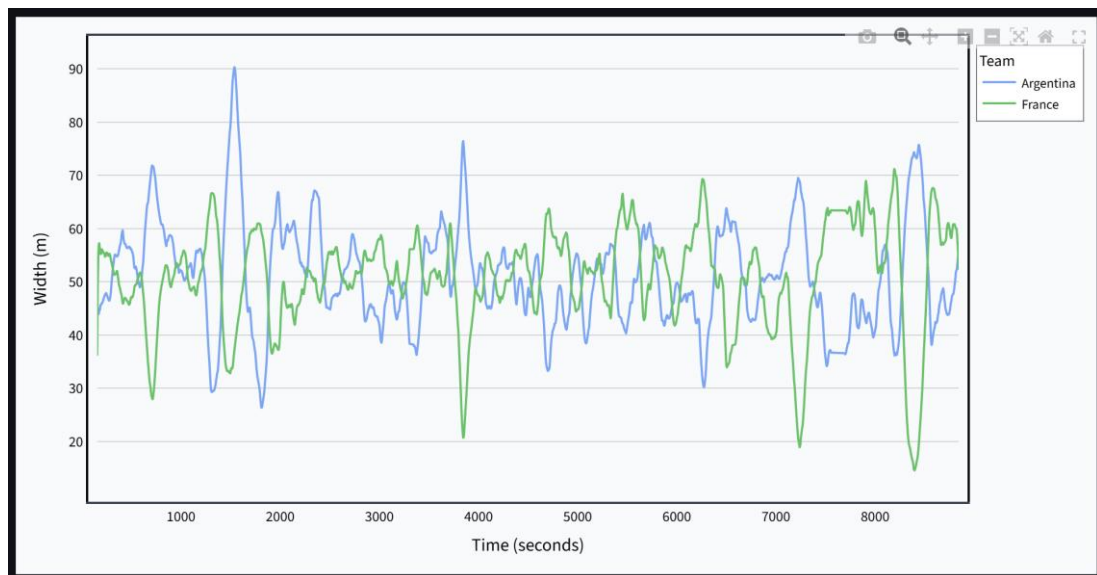


Figure 13: Temporal Line Chart Showing Team Width Variations in the France vs Argentina Match

Figure 13 presents a temporal line chart illustrating variations in team width throughout the France vs Argentina match using positional tracking data. The chart shows continuous fluctuations in horizontal team spacing over time, reflecting dynamic changes in tactical organisation during different phases of play.

The line charts were created using Plotly to visualise metric changes across match timestamps. These visualisations helped analyse how team structures and spatial organisation changed during different phases of play.

The charts also supported comparison between teams by highlighting variations in spatial expansion, compactness, and tactical organisation throughout matches. Filtering components further enabled users to analyse different match segments and metric categories within the system.

3.7.3 Comparative Team Analysis

The dashboard used comparative visualisation methods to analyse spatial differences between teams during matches and across multiple games. This analysis was facilitated by line graphs and bar charts based on calculated spatial metrics.

The visualisations allowed for the comparison of team width, depth, compactness, and convex hull area across matches such as the Netherlands vs Senegal, France vs Argentina, and Germany vs Spain. These comparisons helped analyse how teams organise tactically and behave spatially during gameplay.

Interactive filtering controls were added to the dashboard, enabling users to switch seamlessly between matches, teams, and spatial metrics. This enhancement increased analytical flexibility and facilitated interactive comparisons of football match dynamics within one unified dashboard.

3.7.4 Summary Tables and KPI Metrics

The dashboard integrated summary tables and KPI-style components to present football-tracking data and computed spatial metrics in a structured, visually accessible format. Team summary cards displayed important match-related information, including team names, players tracked, total tracked time, and average team speed for each selected match.

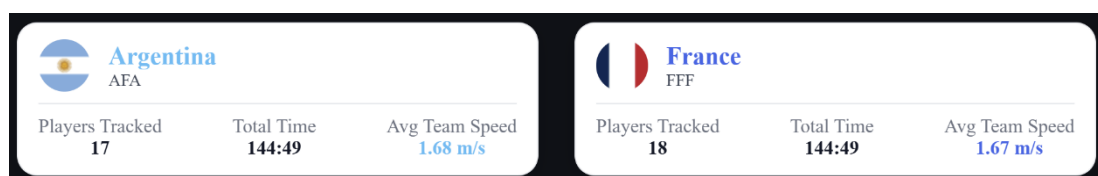


Figure 14: KPI Summary Cards for France and Argentina

Figure 14 presents KPI summary cards displaying key match statistics, including tracked players, match duration, and average team speed for the France vs Argentina match.

Metric summary tables were also included to show processed spatial metrics like width, depth, compactness, and convex hull area for various teams and matches. These tables made it easier to interpret, compare, and organise the analytical results derived from positional tracking data.

Additional dashboard features encompassed automatic insight generation and the ability to download metric summary tables in CSV format. These interactive elements enhanced usability and allowed users to explore, compare, and export football analytics more flexibly within the visual analytics environment.

3.8 Chapter Summary

This chapter detailed the methodology for creating the interactive football analytics dashboard and analysing positional tracking datasets. It covered dataset structure, preprocessing methods, spatial metric calculations, dashboard development, and visualisation techniques used in the study.

Python libraries such as Pandas, NumPy, SciPy, Plotly, and Streamlit were used for data processing, metric calculation, interactive visualisation, and dashboard development. This approach allowed for frame-by-frame spatial analysis and interactive examination of football match dynamics through positional tracking data.

This chapter presents the results and analysis derived from the developed dashboard and the calculated spatial metrics.

Chapter 4 — Results and Analysis

4.1 Introduction

This chapter presents the results and analysis from the interactive football analytics dashboard, which utilises positional tracking data from selected matches. It focuses on assessing spatial metrics, team formations, and tactical patterns through interactive visualisations and spatial analysis techniques.

The chapter analyses three football matches: the Netherlands vs Senegal, France vs Argentina, and Germany vs Spain. It analyses spatial metrics like team width, depth, compactness, centroid position, and convex hull area to investigate differences in team organisation, spatial coverage, and tactical behaviour during different phases of the game.

The findings are presented using pitch-based visuals, temporal line charts, comparison graphs, and summary metrics from the dashboard. This chapter also explores how

interactive visual analytics support tactical interpretation and improve the analysis of complex positional tracking data.

4.2 Netherlands vs Senegal Match Analysis

The analysis of the Netherlands vs Senegal game focused on comparing each team's spatial arrangement and tactical setup throughout play. By utilising positional tracking data, it explored variations in team width, depth, compactness, and overall spatial coverage during different match phases.

The dashboard visualisations allowed for comparison of player positions, team movement patterns, and overall spatial behaviours between the two teams. The match also showed how tracking data can reveal tactical changes and team formations using interactive football analytics methods. These observations provided the foundation for the detailed spatial metric analyses presented in the following subsections.

4.2.1 Team Width Analysis

The team width was analysed to assess how the Netherlands and Senegal used horizontal space during the match. Width was determined by calculating the difference between the highest and lowest x coordinate values for each team at each timestamp. To enhance readability, the dashboard presented this metric using a 120-point rolling average, which smoothed out frame-by-frame noise while maintaining the overall trend.

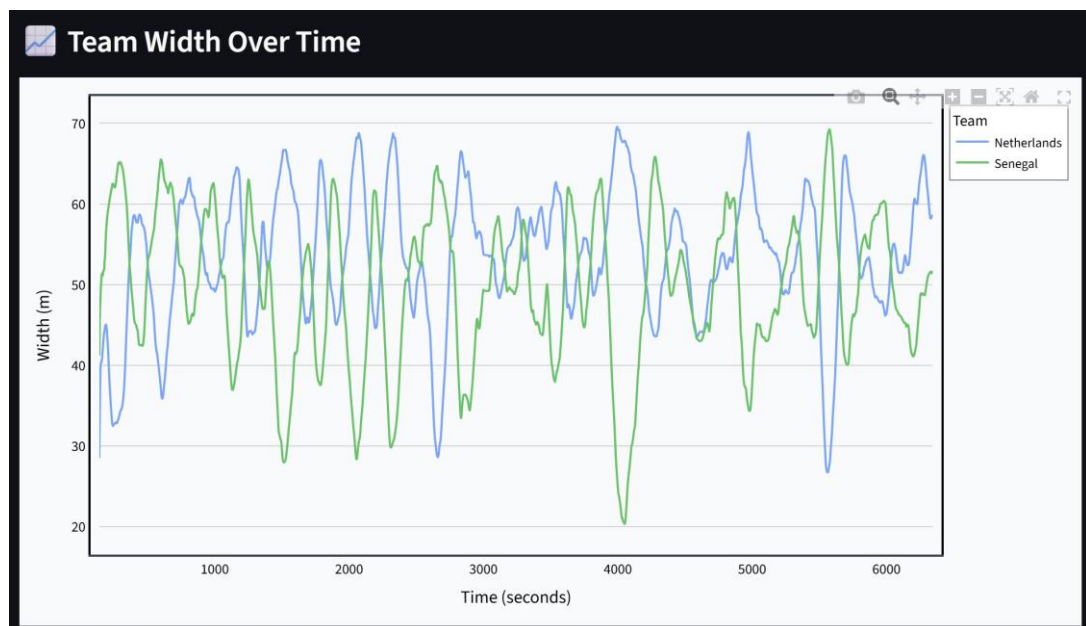


Figure 15: Smoothed Team Width Over Time for Netherlands vs Senegal

Figure 15 illustrates the variation in team width over time for both teams during the Netherlands vs Senegal match. The Netherlands generally maintained wider spatial coverage across several match phases, with width values frequently exceeding 60 m and peaking near 70 m. This indicates broader lateral player positioning and greater use of the full pitch width during attacking and transitional phases.

In contrast, Senegal exhibited more variable width patterns, with multiple sharp contractions where the width fell below 30 m. These moments likely indicate defensive compactness, narrower formations, or quick tactical adjustments. However, Senegal also showed periods of considerable horizontal expansion, achieving width values similar to the Netherlands during attacking phases.

The ongoing oscillation in both lines emphasises the dynamic nature of football team formations. Instead of fixed setups, both teams constantly changed their horizontal spacing based on possession, pressing effort, defensive structure, and transition phases. The smoothed visualisation clearly displayed these overall tactical patterns while reducing frame-by-frame noise from the raw tracking data.

Overall, the width analysis shows how positional tracking data can reveal tactical behaviours related to spatial occupation and team organisation that are not visible through traditional football statistics. These spatial differences reflected how each team adapted its lateral structure during different gameplay situations.

4.2.2 Team Depth Analysis

The team depth was analysed to assess the vertical spacing and structural expansion of both teams during the Netherlands versus Senegal match. It was measured by calculating the difference between the highest and lowest y-coordinate values for each team at every timestamp from the positional tracking data.

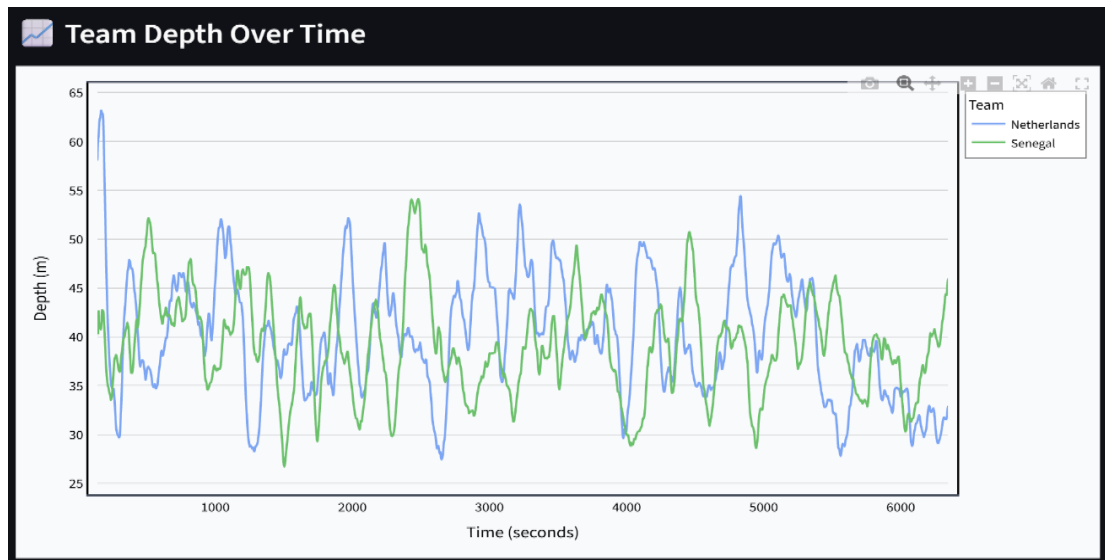


Figure 16: Smoothed Team Depth Over Time for Netherlands vs Senegal

Figure 16 shows continuous fluctuations in team depth throughout the match, indicating dynamic vertical movement and positional adjustments by both teams. The Netherlands generally maintained depth values between approximately 30 m and 55 m across most phases of play, while Senegal displayed depth variations ranging from around 27 m to 54 m.

The Netherlands exhibited multiple periods of greater depth, particularly when values exceeded 50 m. These increased vertical distances indicate moments of wider separation between defensive and attacking units, possibly signifying forward advancement and transitions to attack. Nonetheless, closer vertical organisation was temporarily observed towards the end of the game, reflecting phases of more compact positioning.

Senegal showed relatively balanced depth variations across several periods, but there were notable peaks and drops during the match. Sudden increases in depth reflect phases of vertical expansion during attack sequences, whereas lower depth values indicate tighter defensive organisation with less spacing between players.

The analysis of temporal depth demonstrates how positional tracking data can uncover shifts in team structure and tactical arrangements during a game. The smoothed visualisation enhanced the understanding of overall depth patterns and allowed for a clearer comparison of vertical spatial behaviours between the Netherlands and Senegal throughout the match.

4.2.3 Team Compactness Analysis

Team compactness was assessed to measure how closely players maintained their spatial formation across various phases of the Netherlands vs Senegal game. It was measured by calculating the average Euclidean distance from each player to the team centroid at each time step. Lower values suggest players were positioned more tightly together, whereas higher values indicate wider spatial spread across the field.

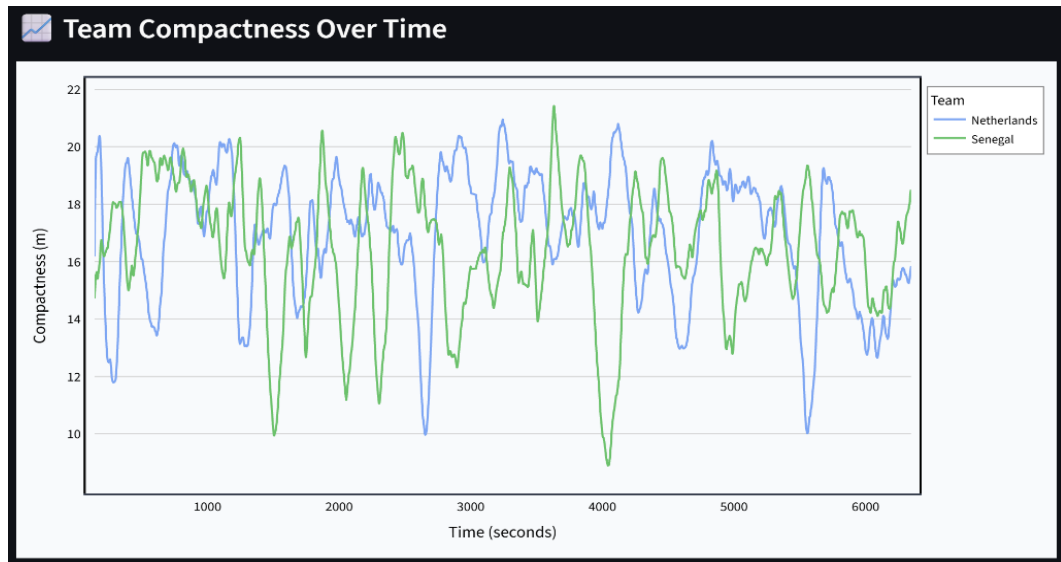


Figure 17: Smoothed Team Compactness Over Time for Netherlands vs Senegal

Figure 17 shows continuous fluctuations in compactness values throughout the match for both teams. The Netherlands generally recorded compactness values between approximately 10 m and 21 m, while Senegal recorded values between approximately 9 m and 21.5 m across different match phases.

During the match, both teams showed noticeable decreases in compactness at certain times. These drops suggest moments when players kept closer together, indicating a more organised, compact defensive setup with shorter gaps between teammates. Conversely, higher compactness values occurred during attacking phases and transitions, when the teams spread out more widely across the field.

Throughout the match, the Netherlands maintained relatively stable compactness patterns, with brief drops around the middle and later stages of play. In contrast, Senegal exhibited more pronounced fluctuations at specific times, reflecting quick shifts from compact defensive formations to broader spatial coverage.

The compactness analysis shows how positional tracking data can uncover differences in team cohesion and tactical setup during a football game. The smoothed temporal visualisation enhanced the understanding of overall spatial patterns and made it easier to compare the team compactness between the Netherlands and Senegal.

4.2.4 Convex Hull Area Analysis

The convex hull area was analysed to assess each team's overall spatial coverage during the Netherlands vs Senegal match. This metric was computed by drawing a convex polygon around player positions at each timestamp with SciPy's ConvexHull() function. The resulting area indicates the amount of pitch space collectively occupied by a team.

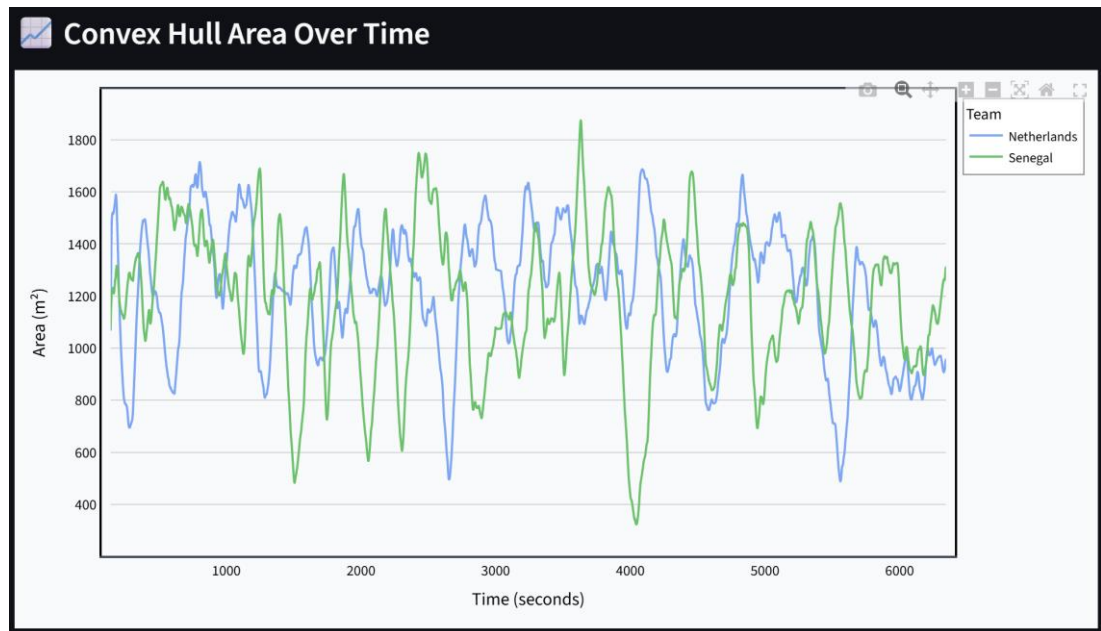


Figure 18: Smoothed Convex Hull Area Over Time for Netherlands vs Senegal

Figure 18 illustrates continuous variation in convex hull area values throughout the match for both teams. The Netherlands generally recorded area values ranging between approximately 500 m² and 1700 m², while Senegal fluctuated between around 330 m² and 1870 m² during different phases of play.

Higher convex hull area values show times when players spread out more across the pitch, indicating broader tactical coverage and greater spatial use. Lower values suggest more compact team formations with less overall pitch coverage. Both teams exhibited repeated patterns of expanding and contracting during attacking and defensive transitions.

The Netherlands maintained relatively stable area fluctuations throughout various match phases, with noticeable decreases during compact defensive periods. Senegal experienced more pronounced fluctuations at specific times, with marked increases and sudden decreases in occupied space, reflecting quick tactical adjustments during the game.

The convex hull analysis demonstrates how positional tracking data can assess team shape, spatial coverage, and tactical organisation during football matches. The smoothed temporal visualisation enhanced understanding of overall spatial occupation trends and facilitated clearer comparisons between the Netherlands and Senegal at various match stages.

4.2.5 Pitch-Based Tactical Visualisation

Pitch-based tactical visualisation was employed to analyse the spatial arrangement and positional structure of both teams in the Netherlands vs Senegal match. This visualisation integrated player positional coordinates, team centroid positions, and convex hull polygons to depict team shape and spatial presence at specific moments during gameplay.

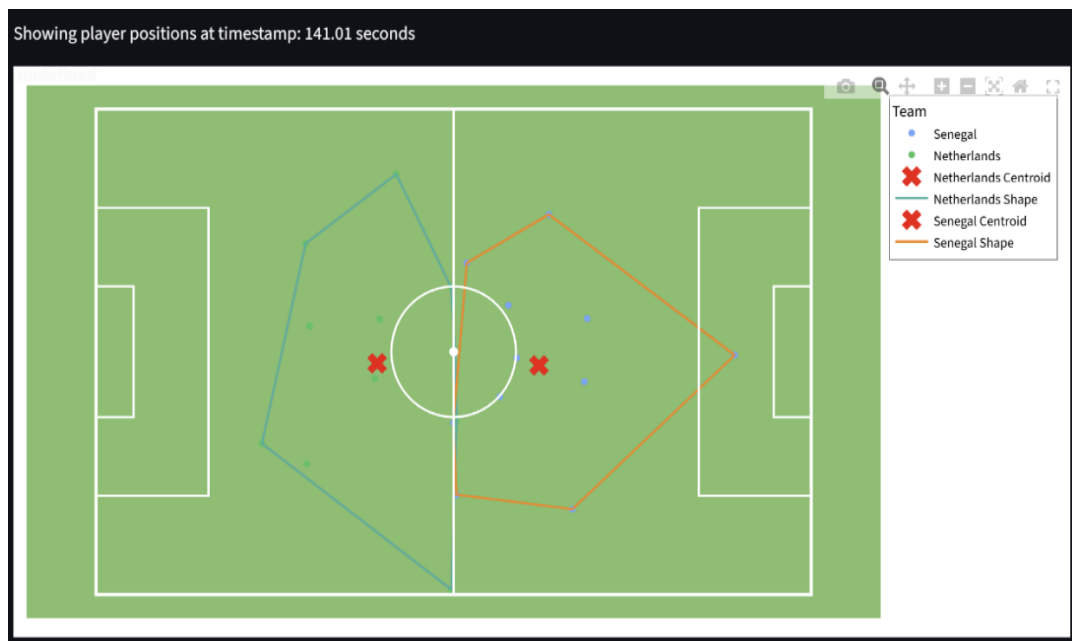


Figure 19: Pitch-Based Tactical Visualisation for Netherlands vs Senegal at 141.01 Seconds

Figure 19 presents a tactical snapshot from approximately 141 seconds into the match, based on positional tracking data. Individual player positions are displayed across the

pitch, while centroid markers indicate the average spatial location of each team at that moment.

The convex hull polygons depict the spatial areas occupied by both teams. The Netherlands showed wider horizontal coverage on the left side of the pitch, with players spread out over a larger area. Senegal's shape was narrower on the opposite side, with a more concentrated formation near the centre-right of the pitch.

The centroid positions reveal that both teams were located near the midfield, suggesting that play was focused around the central area during this phase of the match. Variations in player spacing and polygon structure also underscore differences in tactical organisation, spatial coverage, and overall team shape between the two sides.

This visualisation shows how positional tracking data can yield tactical insights beyond standard football stats. By combining player coordinates, centroid calculations, and convex hull analysis into a unified pitch view, the dashboard makes it easier to understand team organisation and spatial dynamics during live matches.

4.3 Cross-Match Comparative Analysis

A cross-match comparative analysis was performed to investigate differences in spatial organisation and tactical behaviour across three selected matches: Netherlands vs Senegal, France vs Argentina, and Germany vs Spain. The analysis involved comparing average team width, depth, compactness, and convex hull area using bar charts generated from positional tracking data.

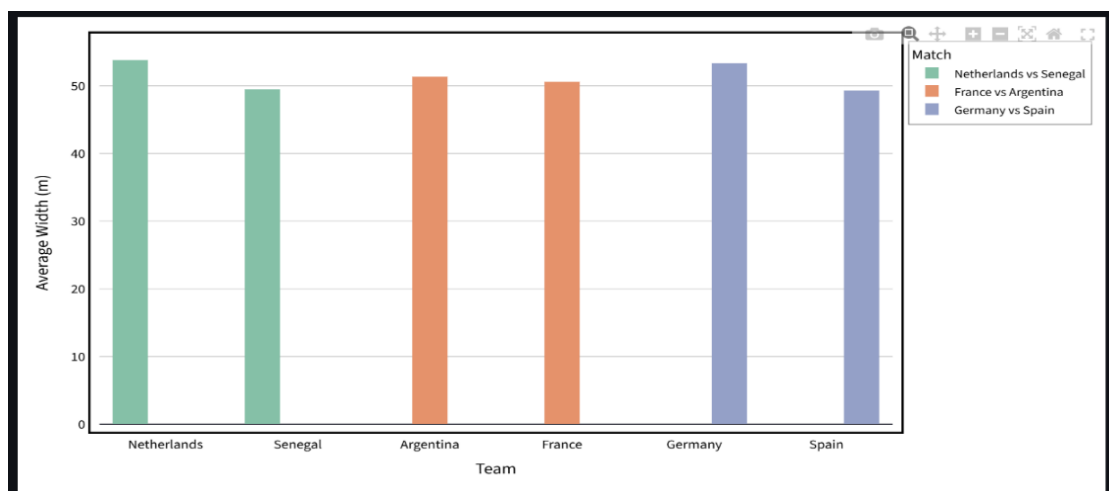


Figure 20: Cross-Match Comparison of Average Team Width

Figure 20 shows the comparison of average team width across all matches. The Netherlands had the highest average width at 53.78 m, with Germany close behind at 53.32 m. Senegal and Spain displayed lower average widths of 49.48 m and 49.28 m, respectively. The broader formations in the Netherlands and Germany indicate greater use of horizontal space and wider player distribution during possession. France and Argentina exhibited more balanced widths of 50.58 m and 51.36 m, reflecting moderate lateral expansion.

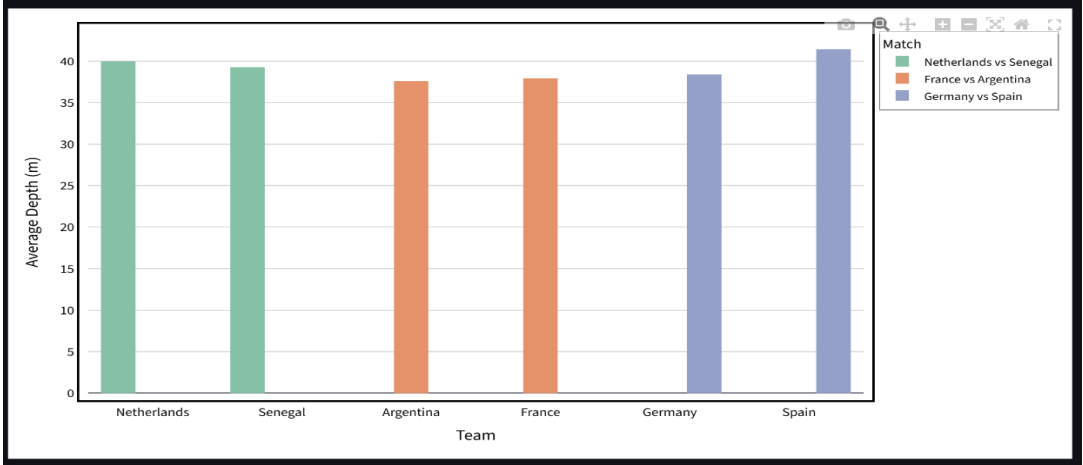


Figure 21: Cross-Match Comparison of Average Team Depth

Figure 21 compares the average team depth values. Spain's average depth was the highest at 41.44 m, whereas Argentina's was the lowest at 37.59 m. The Netherlands and Senegal had similar depth values of 39.98 m and 39.26 m, indicating comparable vertical spacing during the match. Higher depth values typically suggest greater vertical stretching between defensive and attacking units, while lower values imply a more compact vertical formation.

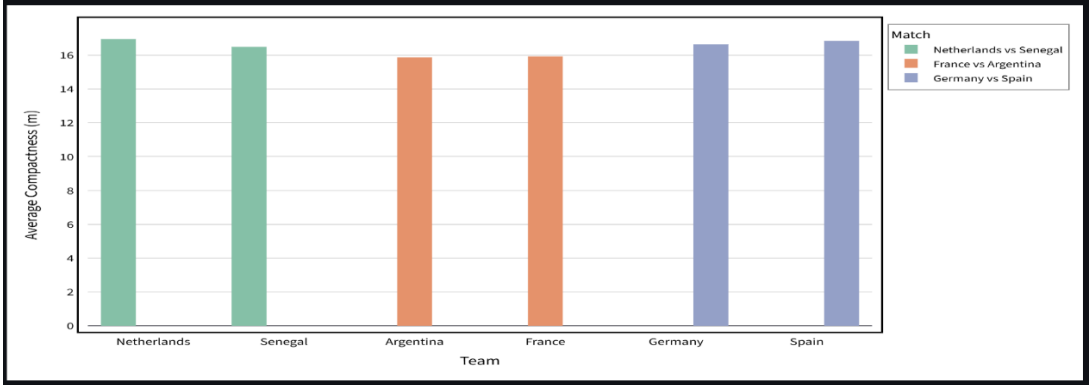


Figure 22: Cross-Match Comparison of Average Team Compactness

Figure 22 compares average compactness values between teams. The Netherlands recorded the highest compactness value at 16.95 m, followed closely by Spain at 16.84 m and Germany at 16.64 m. France and Argentina demonstrated slightly lower compactness values of 15.93 m and 15.87 m, respectively. These differences indicate variations in player proximity around the team centroid. Lower compactness values suggest tighter tactical organisation, while higher values indicate greater player dispersion across the pitch.

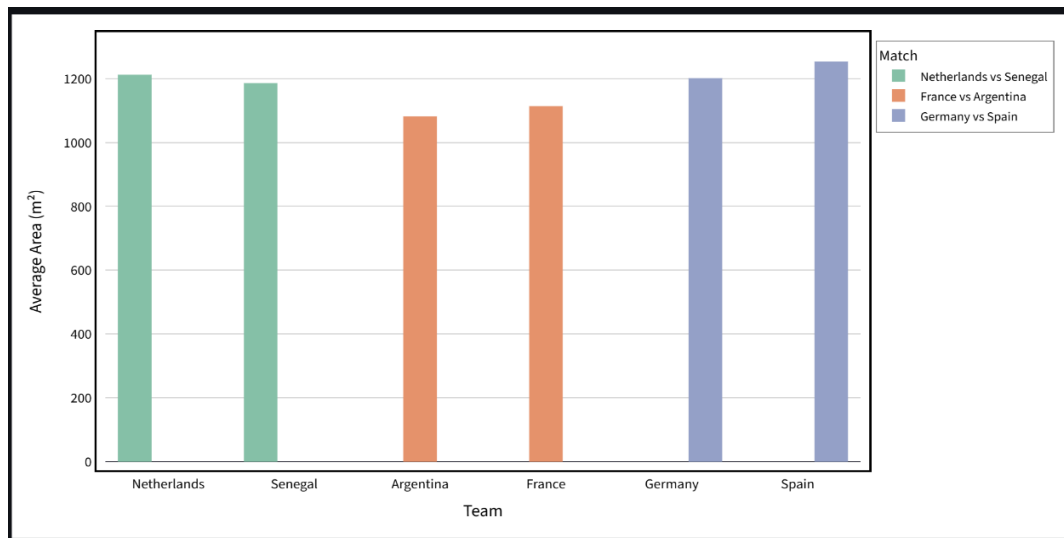


Figure 23: Cross-Match Comparison of Average Convex Hull Area

Figure 23 compares convex hull area values, representing the total spatial coverage of teams during matches. Spain had the largest average area at 1253.41 m², followed by the Netherlands at 1212.17 m² and Germany at 1201.65 m². Argentina exhibited the smallest area at 1081.54 m². The larger convex hull areas for Spain and the Netherlands suggest they employed wider spatial coverage and more expansive positional setups during gameplay phases.

Overall, the cross-match analysis shows how tracking data metrics can uncover important tactical differences between teams. While all teams made dynamic spatial adjustments during matches, the comparative results emphasise variations in horizontal expansion, vertical structure, compactness, and pitch positioning. These findings illustrate the value of positional tracking data and interactive visual analytics in gaining tactical insights that go beyond traditional football statistics.

4.4 Key Findings and Tactical Insights

The analysis of positional tracking data from the selected matches uncovered distinct variations in how teams organised themselves on the field, their tactical setups, and their pitch coverage during gameplay. The spatial metrics showed that football teams consistently adapted their formations in reaction to attacking plays, defensive situations, and positional challenges throughout the matches.

The Netherlands and Germany generally showed higher width values than many other teams, indicating they expanded more horizontally and used more lateral space during possession. Conversely, teams like Senegal and Spain exhibited lower width values at times, reflecting tighter defensive setups and more compact tactics. These differences illustrate how team width reveals tactical approaches to pressing, defensive compactness, and attacking build-up.

Depth analysis further revealed how teams modified their vertical organisation at various points in gameplay. Higher depth values indicated greater separation between defensive and attacking units, typically linked with advancing play and forward movement. On the other hand, lower depth values signified more compact vertical arrangements and tighter defensive formations. The variations seen across matches highlighted the fluidity of football formations, demonstrating that they are constantly evolving rather than fixed in place.

The compactness analysis provided additional understanding of team cohesion and player spacing. Lower compactness values showed players staying closer to the team centroid, indicating organised defensive positioning and smaller gaps between teammates. Higher compactness values indicated players spread out more during attacking phases and transitional moments. These results highlight how positional tracking data can quantitatively reflect tactical organisation changes during matches.

The analysis of convex hull areas highlighted significant differences in the overall spatial occupation among teams. Larger convex hull areas signified more extensive pitch coverage and broader tactical formations, whereas smaller areas indicated compact spatial arrangements and limited pitch presence. Spain and the Netherlands consistently exhibited greater spatial coverage than other teams, implying more expansive positional organisation during various phases of the matches.

Using a combination of width, depth, compactness, and convex hull metrics gives a more complete view of team behaviour than analysing just one metric. For instance, teams with high width and large convex hull areas tend to occupy more space, while those with lower compactness and smaller convex hulls often show a compact defensive setup. Combining multiple spatial metrics allows for a more nuanced assessment of tactical actions and team structure during football matches.

The study also highlights the significance of preprocessing and analysing positional tracking data in football analytics. Since tracking datasets provide continuous spatiotemporal information recorded frame-by-frame, extracting spatial metrics allows complex movement patterns and tactical behaviours to be converted into understandable analytical results. This highlights the growing importance of tracking data processing in modern football performance analysis.

Additionally, the visualisation techniques used throughout the study improved the interpretation of complex spatial data by presenting tactical behaviour through line charts, positional pitch views, comparative graphs, and convex hull representations. These visual outputs supported a clearer understanding of team movement, spatial occupation, and tactical transitions during matches.

Overall, the results show that combining positional tracking data with spatial metrics and visual analytics offers valuable insights into football match dynamics, team organisation, and tactical behaviour, beyond what traditional event-based statistics can provide.

4.5 Chapter Summary

This chapter analysed spatial and tactical patterns in the selected football matches through positional tracking data and derived metric outputs. The analysis highlighted ongoing changes in team formations, player distances, pitch coverage, and positioning during various phases of gameplay.

The findings showed that spatial metrics and tracking data processing can support a deeper understanding of tactical organisation and match dynamics. Visual tools such as temporal charts, pitch views, and comparison graphs help in understanding complex spatiotemporal behaviours during games.

Overall, the chapter demonstrated the value of combining positional tracking analysis with visual analytics techniques to support tactical evaluation in football and comparative match analysis.

Chapter 5 – Conclusion and Future Work

5.1 Introduction

This chapter summarises the key findings obtained from the analysis of football positional tracking data and spatial metrics. It evaluates the achievement of the research objectives, discusses the study's contributions and limitations, and outlines potential directions for future football analytics research.

5.2 Achievement of Research Aim and Objectives

The main objective of the dissertation was to analyse the dynamics of football matches using positional tracking data, spatial metrics, and visualisation. This objective has been accomplished by creating a football analytics framework that can process tracking data, calculate spatial metrics, and create visualisations (Bialkowski et al. 2014).

The positional tracking data was successfully pre-processed for all three football games: Netherlands vs Senegal, France vs Argentina, and Germany vs Spain. The data pre-processing involved organising the data by positional coordinates and timestamps.

Using Python libraries like Pandas, NumPy, and SciPy, several spatial metrics such as team width, depth, compactness, centroid position, and convex hull area were calculated. These metrics provided a quantitative assessment of team organisation, spatial occupation, and tactical behaviour across different play phases (Frencken et al. 2011).

The research further attained its aim of incorporating spatial analysis with the process of visualisation. Line graphs for time, pitch-oriented analysis for tactics, comparative graphs, and metrics were all employed to help analyse tactical structures and spatial behaviour during matches (Stein et al. 2018a).

The results showed that positional tracking data can provide valuable tactical insights beyond standard football statistics. Changes in spatial metrics highlighted differences

in attacking space, defensive compactness, vertical organisation, and overall team structure across matches (Bialkowski et al. 2014).

5.3 Key Contributions of the Study

This study advances football analytics by illustrating how positional tracking data can be converted into understandable indicators of team organisation and tactical behaviour using spatial metrics and visual exploration methods (Fernández and Bornn 2018). It demonstrates how continuous spatiotemporal tracking data can be analysed to assess team movement, spatial positioning, and tactical formations during football matches.

A major contribution of the study was integrating various spatial metrics—such as width, depth, compactness, centroid position, and convex hull area—into a single analytical framework (Duarte et al. 2012). Rather than analysing each metric separately, the study merged them to enable a more comprehensive evaluation of team behaviour and spatial dynamics in different gameplay scenarios.

The study also highlighted the practical importance of visual analytics in football performance analysis. Using temporal charts, pitch-based visualisations, comparative graphs, and summary metrics, the research improved the interpretation of complex tracking datasets and supported a clearer understanding of football match tactics (Stein et al. 2018a).

5.4 Limitations of the Study

However, despite successful data analysis of football positional data, some weaknesses remain. For example, the data used for the analysis came from only three football games, which limited the scope of the comparative analysis.

Most of the research concentrated on analysing data from a team-level spatial perspective and not looking into the performance or passing patterns among individuals or events like shots or tackles (Gyarmati and Hefeeda 2016). Also, many ball-related variables had missing data, making it difficult to analyse ball movement and possession.

Another limitation was the use of smoothed temporal visualisations. Although smoothing improved readability and reduced frame-level noise, some short-term

tactical transitions and quick positional changes might have been less noticeable in the final visuals (Perin et al. 2013).

The study also focused mainly on descriptive spatial analysis instead of predictive modelling or machine learning methods. Consequently, the research prioritised tactical interpretation and visual analysis over automated tactical classification or performance prediction.

5.5 Future Work

Future research could expand this study by including larger datasets of positional tracking from various leagues, tournaments, and seasons. Analysing a wider variety of matches would enable more thorough tactical comparisons and enhance the generalizability of the results (Bialkowski et al. 2014).

Moreover, several other techniques can be incorporated into future technological developments. Some of the other techniques include machine learning, player grouping, passing networks, and automatic tactic pattern identification (Decroos et al. 2019).

Other studies could also use event-based data alongside positional data to provide more context when analysing football tactics. Incorporating player movement data along with pass, shot, and defender data could help better understand attacks and defensive plays in games.

Real-time tracking and analysis, as well as live tactical monitoring systems, could also be explored in future research on football analytics. Such developments may support coaches, analysts, and researchers in analysing tactical behaviour dynamically during matches (Stein et al. 2018b).

5.6 Final Conclusion

This study demonstrated the importance of positional tracking data in analysing football team organisation, spatial behaviour, and tactical dynamics during matches (Gudmundsson and Horton 2017). By integrating spatial metrics with visual analytical methods, the research offered a clearer understanding of how teams expand, contract, and reorganise across various phases of play.

The findings indicate that spatial metrics like width, depth, compactness, and convex hull area can enhance tactical understanding by converting complex tracking data into meaningful indicators of team behaviour (Frencken et al. 2011). Additionally, using comparative visualisations can help interpret spatial patterns and tactical formations during football matches.

The study further highlighted the practical benefits of interactive visual analytics for football performance. The dashboard allowed users to analyse positional tracking data with tools such as temporal charts, pitch visualisations, KPI summaries, and tactical comparisons, making it easier to interpret complex football datasets.

Although the research focused primarily on descriptive spatial analysis using a limited number of matches, the results highlight the potential of combining positional tracking data with interactive analytical systems for tactical evaluation and football performance analysis. Future work could incorporate larger datasets, machine learning techniques, event-based football data, and real-time tactical monitoring approaches.

Overall, this dissertation successfully achieved its research aim by developing a football-focused visual analytics system that processes positional tracking data, computes spatial metrics, and supports tactical interpretation through interactive visualisation techniques. The study demonstrates how analytical and visual approaches can contribute to improved understanding of football match dynamics and modern football analytics.

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Appendix

Appendix A: Python Data Preprocessing Code

```
import pandas as pd
import numpy as np

df = pd.read_csv("tracking_data.csv")

df = df.dropna(subset=['x', 'y'])

df['timestamp_s'] =
pd.to_numeric(df['timestamp_s'])
df = df.sort_values(by='timestamp_s')
```

Appendix B: Team Width Calculation

```
def calculate_team_width(team_df):
    return team_df['x'].max() -
team_df['x'].min()
```

Appendix C: Team Depth Calculation

```
def calculate_team_depth(team_df):
    return team_df['y'].max() -
team_df['y'].min()
```

Appendix D: Team Compactness Calculation

```
def calculate_compactness(team_df):
    centroid_x = team_df['x'].mean()
    centroid_y = team_df['y'].mean()

    distances = np.sqrt(
        (team_df['x'] - centroid_x)**2
+       (team_df['y'] - centroid_y)**2
    )

    return distances.mean()
```

Appendix E: Convex Hull Area Calculation

```
from scipy.spatial import
ConvexHull
points = team_df[['x', 'y']].values

hull = ConvexHull(points)

area = hull.volume
```

Appendix F: Team Centroid Calculation

```
def calculate_centroid(team_df):
    centroid_x = team_df['x'].mean()
    centroid_y = team_df['y'].mean()

    return centroid_x, centroid_y
```

Appendix G: Streamlit Dashboard Sidebar Controls

```
selected_match = st.sidebar.selectbox(
    "Select Match",
    matches
)

selected_metric = st.sidebar.selectbox(
    "Select Metric",
    metrics
)

selected_period = st.sidebar.selectbox(
    "Select Period",
    ["Full Match", "First Half", "Second Half"]
)
```

Appendix H: KPI Summary Cards

```
col1, col2, col3 = st.columns(3)

col1.metric("Players Tracked",
players_count)
col2.metric("Total Time", total_time)

col3.metric("Average Team Speed", avg_speed)
```

Appendix I: Temporal Line Chart Visualisation

```
fig = px.line(
    metric_df,
    x='timestamp_s',
    y='width',

    color='team_name'
)

st.plotly_chart(fig)
```

Appendix J: Pitch-Based Positional Visualisation

```
fig = px.scatter(  
    frame_df,  
    x='x',  
    y='y',  
  
    color='team_name'  
  
st.plotly_chart(fig)
```

Appendix K: Convex Hull Pitch Visualisation

```
hull = ConvexHull(points)  
  
for simplex in hull.simplices:  
    plt.plot(  
        points[simplex, 0],  
        points[simplex, 1]  
    )
```

Appendix L: Match Comparison Visualisation

```
fig = px.bar(  
    comparison_df,  
    x='match',  
    y='compactness',  
    color='team'  
  
)  
  
st.plotly_chart(fig)
```